

00 – Databases

Organisation, People & Schedule

Jérôme Landré – jerome.landre@ju.se

OUTLINE

- 0) Who am I?
- 1) Goal of the course
- 2) ILOs (Intended Learning Outcomes)
- 3) Organisation
- 4) Presentation of the team
- 5) Grades
- 6) Rules
- 7) Course evaluation / Changes this year
- 8) How to succeed in this course?

0) WHO AM I?

Jérôme Landré
Assistant Professor
Ph.D in Image Processing
Born in 1972
Car (number/license) plate collector



Career

Jönköping University (4 years)
Université de Reims Champagne-Ardenne (16 years)
Université de Bourgogne (10 years)

Research

Image Processing and Computer Vision
Applications to medical images, satellite images, patrimonial images...

Teaching

Web development (HTML, CSS, JS, NodeJS)
Operating Systems (Linux & Windows)
Databases (SQL, NoSQL)

STOP

Didn't you remark anything weird in the previous slide?

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AND THE QUESTION IS



?

BECAUSE, AT FIRST, IT'S



BECAUSE IT'S

**A WAY TO KEEP YOUR ATTENTION
DURING THE LECTURES...**

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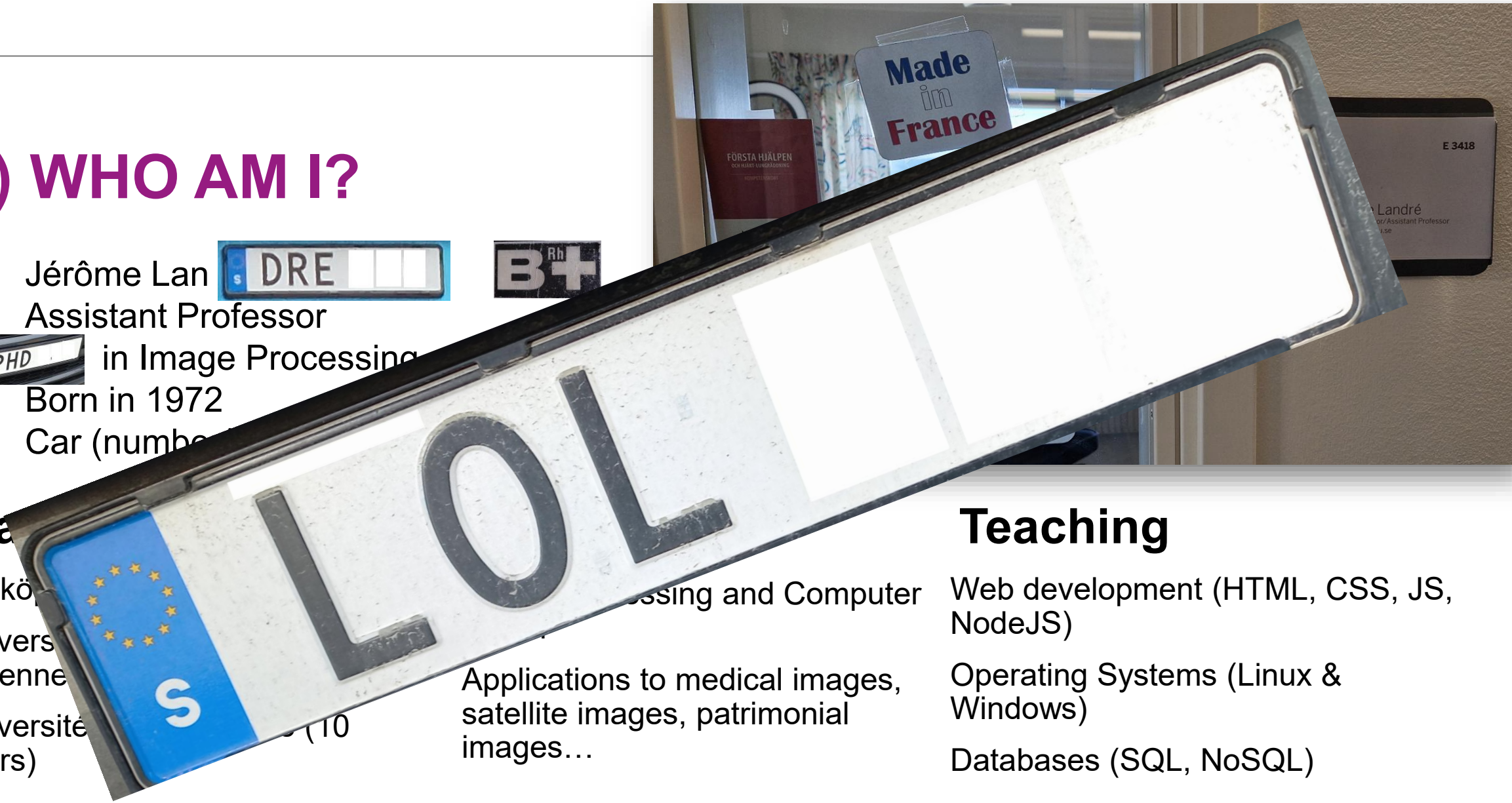
Ca

Jönkö

Univers

Ardenne

Université
(10
years)



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**Made
in
France**

FÖRSTA HJÄLPEN
OCH HJÄRT-LUNGRÄDDNING

KOMPETENSOMRÅDET

HLR  rådet



Jag trivs i Sverige

E 3418

Jérôme Landré

Universitetslektor/ Assistant Professor
jerome.landre@ju.se

1) GOAL OF THE COURSE

Learn, in **ENGLISH**, the basics of databases with a focus on Relational Databases:

- 1) Modelisation/Design:
 - **Entity-Relationship Diagram (ERD)**
 - Entities,
 - Attributes,
 - Relationships,
 - Cardinalities.
- 2) **Structured Query Language (SQL)**
 - **SELECT ... FROM ...**
 - **INSERT INTO ...**
 - **UPDATE ... SET ... WHERE ...**
 - **DELETE FROM ... WHERE ...**
- 3) **Database Administration/Usage:**
 - Database **servers**
 - Database **accounts** and **privileges**
 - **Software** and **code examples**
- 4) And several other database systems:
 - **Graph Databases,**
 - **NoSQL,**
 - ...



To increase your **engagement**, I will include slides with **music, tips and tricks, fun facts** about **professional** and **personal** life as well as some French language...

TIPS AND TRICKS

POSITIVE ATTITUDE CAN CHANGE LIFE

Syed Muhammad Sajjad Kabir

Chairman and Assistant Professor, Department of Psychology

BE POSITIVE!

*Remember that **YOU** control your attitude:*

Use a more positive vocabulary:

Be thankful:

Smile:

Understand that we all need and deserve time to recharge:

Follow a healthy lifestyle:

Be open to humor: Give yourself permission to smile or laugh, especially during difficult times. Seek humor in everyday happenings. When you can laugh at life, you feel less stressed.

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BE POSITIVE!

Be grateful or have gratitude in what you have today. Learn to utilize what you have in your hands and make use of it in the most constructive way.

When you fall, you must understand that everyone falls no matter how great or how stable they may look. When everything seems to fall apart, tell yourself that if you could get out of this mud of sufferings, you could be one of the greatest persons on earth.

Positive attitudes lead to happiness and success and can change your whole life.

TIPS AND TRICKS

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When you fall, you must understand that they may look. When everything seems to run up and down, mud of sufferings, you could be one of the greatest persons on earth.

We've gotta hold on to what we've got
It doesn't make a difference if we make it or not

...
We'll give it a shot

<https://www.youtube.com/watch?v=IDK9QqIzhwk>

Positive attitudes lead to happiness and success and can change your whole life.

2) ILOS (INTENDED LEARNING OUTCOMES)

Intended Learning Outcomes (ILO)

On completion of the course the student shall:

Knowledge and understanding

- demonstrate comprehension of a relational database (function, architecture, and limits)
- display knowledge of fundamental database design principles
- display knowledge of transaction management, indexing, and file organization within databases
- demonstrate comprehension of the main differences between relational and non-relational databases

Skills and abilities

- demonstrate the ability to design an ER model from a provided business description
- demonstrate the ability to apply the first three normalization forms
- demonstrate the ability to formulate basic and advanced SQL queries
- demonstrate the ability to interpret SQL expressions and comprehend the obtained results
- demonstrate the ability to understand and use No-SQL databases

4) ORGANISATION

- Canvas pages of the course:
 - TDRK16: <https://ju.instructure.com/courses/16159>
- 100+ students from diverse programs, from different countries

4) ORGANISATION

- Learn many concepts in a progressive way:
 - **Relational Databases** basics: Why? What? How?
 - Database **model**: conceptual, logical and physical
 - Database **optimization: normal forms** 1-NF, 2-NF, 3-NF, ...
 - **Structured Query Language (SQL)** using JTH MS SQL Server
 - Database **administration** using MariaDB (within Docker)
 - Database **usage** (with code examples) using Node.js (within Docker)
 - Other types of **modern** databases: **graph** databases, **NoSQL**...

4) ORGANISATION

- Per week, there are:
 - **One** 1:45 hours lecture (10 in total)
 - **One** 1:45 hours interactive exercise sessions in an Active Learning Classroom, ALC (9 in total)
 - **One** 3:45 hours lab session (7 in total)
- Online schedule made by Daniel LUNDQVIST, tack!
<https://cloud.timeedit.net/ju/web/staff/ri10w30Xn7jZ0YQ84QQ0Zv94tQ1Y8m20Z9Yk75od0n7j1650599y1QAZA0Q5c79A7w27Q2666qZ9nEcZ0F9Z0pE3tu26Q1947QEBEE0EC2C.phtml>
- Be careful:
 - Written **examination** scheduled on Wednesday, May 27th, 2026
 - **Several changes** can occur during the course due to constraints on the teachers, rooms or, other unpredictable events
- You will receive a notification on Canvas if there is any change

4) ORGANISATION

- **WORKSHOPS**

- Participation is **individual** (no need to be two students)
- **Small progressive exercises** to apply **the theory** with discussions with other students in an Active Learning Classroom (ALC)

- **LABS**

- 5 labs for TDRK16 Databases
- You must do **all the labs** and **present them** to the lab assistants
- **2** students per lab group
- When you present **ALL** the students in the group must be **PRESENT**
- **Small progressive exercises** to apply **the theory** to a case study

4) PRESENTATION OF THE TEAM



?

4) PRESENTATION OF THE TEAM



Jérôme

Coordinator,
Lecturer and
Lab assistant



Kyla

Previous student
of the course and
Lab assistant



Abdulqaadir

Previous student
of the course and
Lab assistant



Arvid

Previous student of
the course and Lab
assistant

5) GRADES

- The course is graded 5, 4, 3 or Fail (U).
- It is divided in two parts:
 - **Theory** (graded 5, 4, 3, Fail U):
 - Written examination mainly
 - **Practice** (labs, graded Pass G, Fail U):
 - Labs presentation

TRDK16 Databases

Examination and grades

The course is graded 5, 4, 3 or U.

Registration of examination:

Name of the Test	Value	Grading
Examination ¹	5 credits	5/4/3/U
Assignment	2.5 credits	G/U

¹Determines the final grade of the course, which is issued only when all course units have been passed.

Written examination

(don't forget to
register for it!)

Databases 7.5 HP- TDRK16

TDRK16/2601 5 HP (2601 Examination)

Wednesday 27 May time 09:00 to 13:00

The registration will open 2026-04-27

Type of exam: Inspera – Digital exam **Anonymous**

Room: **Campus Jönköping**

Free text: Read and find the rules for exams here:

<https://ju.se/student/en/studies/examination.html>

6) RULES

- Your role as a student:
 - be **engaged**, **active** and **interested** by the course:
 - actively attend the lectures to get the building blocks of knowledge necessary to understand the different concepts
 - actively attend all the sessions to have the theory and practice on use cases
 - work and ask for help and feedback during the labs to apply the concepts to real-life professional situations
 - Ask questions to the teachers, ask for help when needed
 - Take into account teachers' feedback to improve the assessments
 - Give feedback on advantages and drawbacks of this course to the coordinator
 - Be **polite** and **respectful**
 - Don't **speak too loud** during lecture/workshop/lab sessions
 - Be **POSITIVE!**

6) RULES

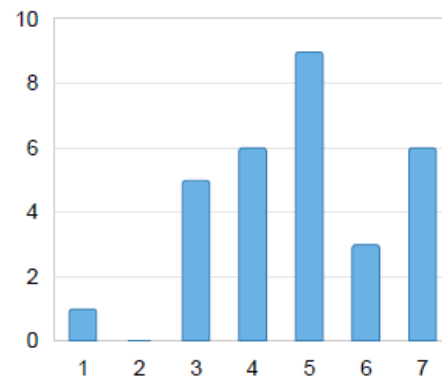
- Our role as teachers:
 - **Facilitate** (from Björn Praestgaard-Larsen) students' learning
 - **Make** the students discover and explore the concepts on their own (with questions and feedback)
 - **Create** a good pedagogical environment for students
 - Be **inclusive** to include every student
 - Be **polite** and **respectful**
 - Be **POSITIVE!**
- Contact:
 - To contact me, send me an **email**: jerome.landre@ju.se
 - I do **NOT** answer emails during weekend days

7) COURSE EVALUATION / CHANGES THIS YEAR

- From two years ago:

My opinion of the course as a whole.
1 = Not at all satisfied
7 = Very satisfied

My opinion of the course as a whole. 1 = Not at all satisfied 7 = Very satisfied	Number of responses
1	1 (3.3%)
2	0 (0.0%)
3	5 (16.7%)
4	6 (20.0%)
5	9 (30.0%)
6	3 (10.0%)
7	6 (20.0%)
Total	30 (100.0%)

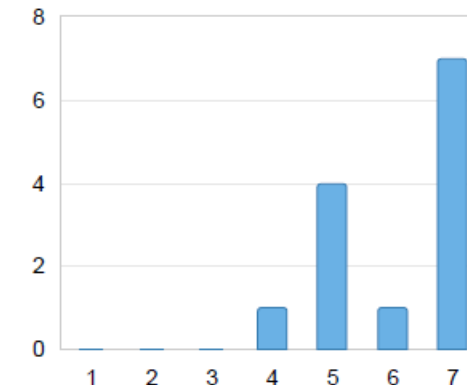


● My opinion of the course as a whole....

- From last year:

My opinion of the course as a whole.
1 = Not at all satisfied
7 = Very satisfied

My opinion of the course as a whole. 1 = Not at all satisfied 7 = Very satisfied	Number of responses
1	0 (0.0%)
2	0 (0.0%)
3	0 (0.0%)
4	1 (7.7%)
5	4 (30.8%)
6	1 (7.7%)
7	7 (50.8%)
Total	13 (100.0%)



● My opinion of the course as a whole....

100+ students!

7) COURSE EVALUATION / CHANGES THIS YEAR

- From two years ago **course evaluation**:

More lab assistants perhaps and all the labs should be placed after the lecture/workshops

I think if teacher makes the groups will make the students to give their best

The Labs where quite hard to understand at somebody points, and 2 teachers would have been better per classroom at labs

Find software that will work on everyone's computer.

I dont think us IT infrastructure students should have this course as it is right now, as it contains very little about data administration. Comparing previous exams in dataadministration compared to this course, I feel that I havent learned much at all about the administration part. Also the labs does not reflect the dataadministration part as it basically only focuses on the databases and how to build them, not administrating it.

It is very difficult to perform the course from distance which is not good.. the pdfs in and of itself are not very easy to understand if you haven't been at the lectures before. A great improvement would be if the lectures were recorded and uploaded to canvas

7) COURSE EVALUATION / CHANGES THIS YEAR

- From last year's **course evaluation**:

More lab assistants perhaps and all the labs should be placed after the lecture/workshops

I think if teacher met with students will make the students to give their best

Good teacher

The teacher Jerome explains well and responds very quickly to questions sent to him through email.

Find software that will work on everyone's computer

Great and encouraging primary lecturer

I don't think our IT infrastructure is good enough right now, as it contains very little about data administration. Comparing previous exams in data administration compared to this course, I feel that I haven't learned much at all about the administration part. Also the labs do not reflect the data administration part as it basically only focuses on the databases and how to build them, not administering it.

I thought overall the lectures and teachers were very good!

It is very difficult to perform the course from distance which is not good. The parts of itself are not very easy to understand if you haven't been at the lectures before. A great improvement would be if the lectures were recorded and uploaded to canvas

7) COURSE EVALUATION / CHANGES THIS YEAR

- From last year **course evaluation**:

What about the course do you think is most important to change or improve?

Azure studio

I think some improvement might be needed with accessing the mssql server.

I would have liked to have more MongoDB, JSON and Graph database lectures. The lectures about Relational Databases were well structured, with both theory and plenty of demonstrations. Whereas with MongoDB and Graph databases, we had very few lectures and demonstrations.

I think it would be beneficial to include a weekly plan in the study guide where the students can see what the teacher is going to go through each week. I also believe it's better if the teacher uploaded the lecture slides in advance so the students can prepare for the lecture

7) COURSE EVALUATION / CHANGES THIS YEAR

- From last year **course evaluation**:

Overall i was very pleased with the course it was fun and i learned alot.

Overall, the lectures were very interesting, the labs were fun and the overall teaching method was pedagogical. The teacher put in the time to explain core concepts, whilst also giving us the freedom to explore our own interests(such as the lab where you could choose your own database topic to design in an ER-diagram)

The whiteboard notes and examination examples

The combination between lectures and workshops is a good way to not only learn the theory of the course, but to use it in a practical environment. The labs ensured that I understand the content of the syllabus and prepare me for the exam.

Jerome is a very nice and funny teacher and he explains on the whiteboard very well. This contributed to me understanding the course

7) COURSE EVALUATION / CHANGES THIS YEAR

So, what's



this year?

7) COURSE EVALUATION / CHANGES THIS YEAR

På svenska:



7) COURSE EVALUATION / CHANGES THIS YEAR

- There is a problem about the **the tool** used to connect to MSSQL server:
 - I have changed to **Visual Studio Code + SQL Server (mssql)** extension from Microsoft
 - And I have a **plan B** in case there are problems
- There is a **waiting queue** in the labs:
 - There are four lab assistants this year to help you!
 - You must present **ALL THE LABS** as a group of **2 students**,
 - You can ask for **help and feedback** during the labs
- There are only a **few exercises** on NoSQL databases:
 - I will add **more exercises** specifically on this topic

8) HOW TO SUCCEED IN THIS COURSE?

- **Come** to the lectures!
- **Participate** in the exercise sessions!
- Do the **labs**, ask **questions**!
- Use AI if you want BUT understand the answer
- (Cite your sources to avoid plagiarism!!!)

Read, read and read again...

8) HOW TO SUCCEED IN THIS COURSE?

REMEMBER, it's all about



Source: <https://news.stanford.edu/stories/2018/06/four-ways-human-mind-shapes-reality>

8) HOW TO SUCCEED IN THIS COURSE?

You are the



8) HOW TO SUCCEED IN THIS COURSE?

You must find your own



8) HOW TO SUCCEED IN THIS COURSE?

On the road to success, you are your own



8) HOW TO SUCCEED IN THIS COURSE?

Of course the teachers will help you, but we are only facilitators,

YOU DO THE



8) HOW TO SUCCEED IN THIS COURSE?

If you lack knowledge, just fill the



8) HOW TO SUCCEED IN THIS COURSE?

So STAY

READ THE

HAVE

AND BE



SO, WELCOME ON BOARD:

DATABASES

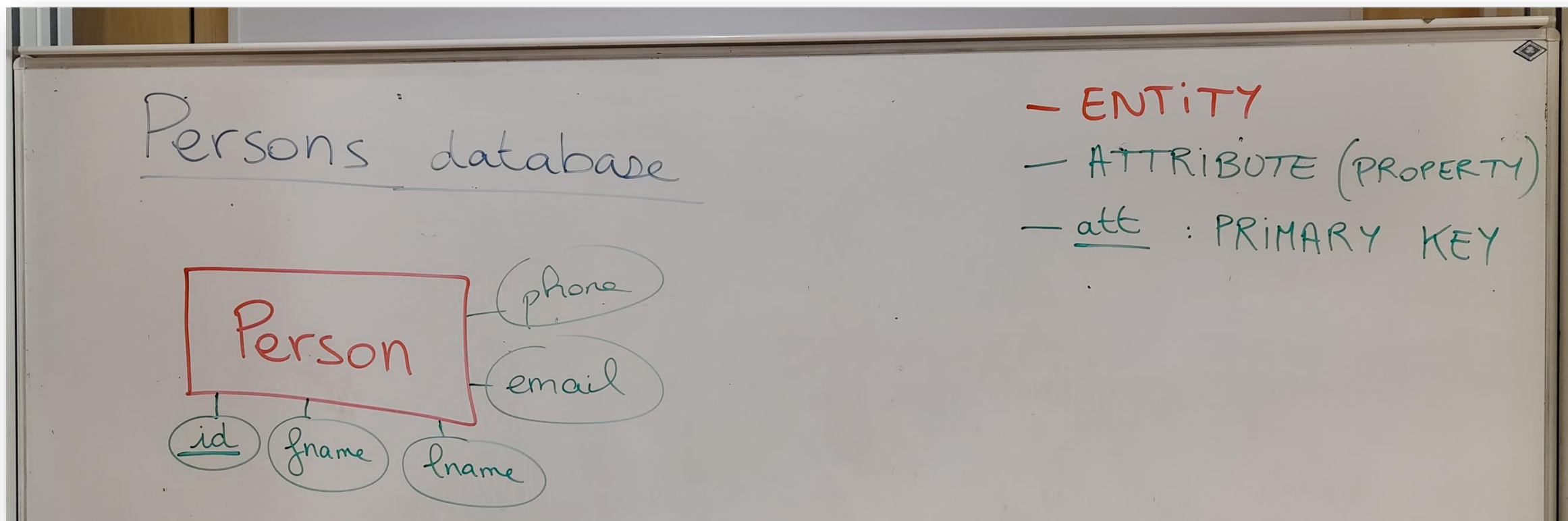
REMEMBER:

DATA IS POWER!

ANY QUESTIONS?

LET'S PLAY WITH MARIADB

I want to create a database to save persons information. What information do I need to save the data? Let's draw it on the white board together...



LET'S PLAY WITH MARIADB

You need to install Docker (<https://www.docker.com/>), once done, open a terminal and run the commands:

PS C:\Users\jerome> docker --version

Docker version 29.2.1, build a5c7197

PS C:\Users\jerome> docker pull mariadb:latest

latest: Pulling from library/mariadb

...

docker.io/library/mariadb:latest

PS C:\Users\jerome> docker run --name my-mariadb -e

MYSQL_ROOT_PASSWORD=my-pw-1234 -d -p 3306:3306 mariadb:latest

PS C:\Users\jerome> docker exec -it my-mariadb bash

LET'S PLAY WITH MARIADB

Linux Terminal { root@de5afbadd338:/# mariadb -u root -p

Enter password:

Welcome to the MariaDB monitor. Commands end with ; or \g.

Your MariaDB connection id is 5

Server version: 12.2.2-MariaDB-ubu2404 mariadb.org binary distribution

MariaDB [(none)]> SHOW DATABASES;

MariaDB [(none)]> CREATE DATABASE Persons;

MariaDB [(none)]> USE Persons;

MariaDB [Persons] >

LET'S PLAY WITH MARIADB

MariaDB [Persons]> CREATE TABLE Person (id INT PRIMARY KEY AUTO_INCREMENT, fname VARCHAR(60), lname VARCHAR(90), email VARCHAR(100), phone VARCHAR(20));

MariaDB [Persons]> INSERT INTO Person (fname, lname, email, phone) VALUES ('Jerome', 'Landre', 'jerome.landre@ju.se', '+46 707070707');

MariaDB [Persons]> INSERT INTO Person (fname, lname, email, phone) VALUES ('Mira', 'Pop', 'mira.pop@ju.se', '+46 606060606');

MariaDB [Persons]> SELECT * FROM Person;

```
+----+-----+-----+-----+-----+
| id | fname | lname | email           | phone       |
+----+-----+-----+-----+-----+
| 1  | Jerome | Landre | jerome.landre@ju.se | +46 707070707 |
| 2  | Mira  | Pop   | mira.pop@ju.se    | +46 606060606 |
+----+-----+-----+-----+-----+
2 rows in set (0.001 sec)
```

MariaDB [Persons]>

INSERT INTO: **C**REATE

SELECT: **R**EAD

LET'S PLAY WITH MARIADB

```
MariaDB [Persons]> UPDATE Person SET lname='Landré' WHERE id=1;  
Query OK, 1 row affected (0.004 sec)  
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [Persons]> SELECT * FROM Person;
```

id	fname	lname	email	phone
1	Jerome	Landré	jerome.landre@ju.se	+46 707070707
2	Mira	Pop	mira.pop@ju.se	+46 606060606

```
2 rows in set (0.000 sec)
```

```
MariaDB [Persons]>
```

UPDATE: **U**PDATe

LET'S PLAY WITH MARIADB

MariaDB [Persons]> DELETE FROM Person WHERE id=2;
Query OK, 1 row affected (0.002 sec)

DELETE: **DELETE**

MariaDB [Persons]> SELECT * FROM Person;

id	fname	lname	email	phone
1	Jerome	Landré	jerome.landre@ju.se	+46 707070707

1 row in set (0.000 sec)

MariaDB [Persons]> quit

Bye

root@55cd4006e30d:/# exit

exit

PS C:\Users\lanjer> docker stop my-mariadb

my-mariadb

CREATE, READ, UPDATE, **DELETE**:
CRUD

LET'S PLAY WITH MARIADB

PS C:\Users\lanjer> docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
de5afbadd338	mariadb:latest	"docker-entrypoint.s..."	44 minutes ago	Up 43 minutes	0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp	my-mariadb

PS C:\Users\lanjer> docker stop my-mariadb

PS C:\Users\lanjer> docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
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PS C:\Users\lanjer>

ANY QUESTIONS?